Case Study on Design of Test Cases and a Maintenance Plan for a Hospital Appointment Booking Application

Aim

To design test cases and a maintenance plan for a sample application, using a Hospital Appointment Booking Application as the reference case.

Objective

- Identify key modules that require testing.
- Design test cases covering the core booking flow.
- List the relevant testing types.
- Prepare a maintenance plan for post-release support.

Selected Software Application

Hospital Appointment Booking Application

Overview

A Hospital Appointment Booking Application allows patients to search doctors, check available time slots, book appointments, and make consultation payments online, while hospital staff manage doctor schedules and patient records.

Functional Requirements

- Patient Module: Register/Login, Search Doctors, Book/Cancel Appointment, Payment
- Doctor Module: Manage Availability, View Appointments, Update Status
- Admin Module: Manage Doctors, Manage Departments, Reports
- Notification Module: Confirmation, Reminders, Cancellation Alerts

Test Case Design

Test Case	Expected Result
Login	Successful Login
Invalid Password	Error Message
Book Appointment	Appointment Confirmed
Double Booking Attempt	Slot Unavailable Shown
Payment	Payment Successful

Testing Types

- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing (UAT)

Maintenance Plan

Maintenance Type	Example
Corrective	Fixing duplicate slot booking bug
Adaptive	Updating payment gateway API
Perfective	Adding SMS appointment reminders
Preventive	Refactoring slot-locking logic

Maintenance Activities

- Bug Fixes
- Security Updates
- Feature Enhancements
- Performance Optimization

Results

- Test cases helped identify booking-conflict and payment-failure scenarios before release.
- A structured maintenance plan reduced downtime risk for the application.
- Regular maintenance activities keep the system secure and reliable.

Conclusion

The case study shows that well-designed test cases and a structured maintenance plan are essential for a reliable Hospital Appointment Booking Application. Thorough testing of the booking and payment flow, combined with regular maintenance, ensures consistent uptime and a dependable user experience after go-live.